Utilization of HSS Grinding Waste and Mild Steel for the Development of Sustainable Hybrid Composites

B.Tech Project (BTP-2)

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Problem Statement

- HSS grinding operations produce metallic waste, and this waste is not fully used even though it is very hard and has useful material properties.
- The fine grinding waste can be hard to handle and to reuse effectively, so we need ways to use this waste.
- BFS can be mixed separately with HSS grinding waste and with mild steel to create hybrid composite materials that are more stable and have useful properties.



Fig 1: HSS Grinding Powder (Reinforcement Material)

Objectives

- To develop hybrid composite systems by using HSS grinding waste and mild steel, each mixed with BFS.
- To study how the mix of materials and the way they are processed affect the structural features of the composites.
- To evaluate composite performance through density measurement using Archimedes' principle.
- To obtain good binding of atoms together and even mixing by using PVA as a binder.
- To analyse the composites by X-ray diffraction (XRD), scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS).
- To investigate the compression performance of the composites and to correlate this with their composition and internal structure.
- To investigate their brittleness, how they break, and to find a combination with better stability that is useful for real engineering projects.

Methodology

Step 1: Material Collection

- HSS powder, EAF III, BFS collected
- PVA used as binder

Step 2: Mixing

- EAF III + HSS (5:1 ratio)
- PVA added for bonding
- Uniform mixing ensured

Step 3: Pellet Formation

- Pressed into disc shape
- Prepared for heat treatment

Step 4: Heat Treatment

- Heated at 950 ℃ for 2 hours



Fig 2: Hydraulic press used for pellet formation

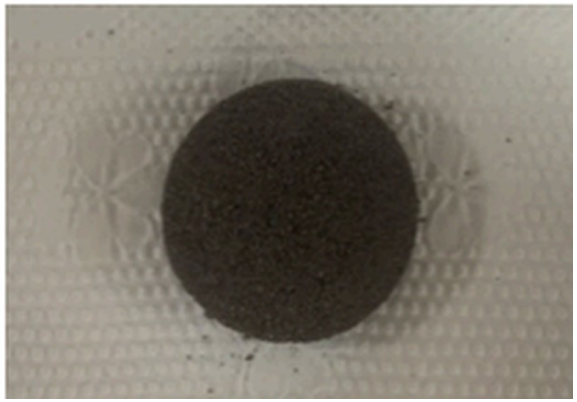


Fig 3: EAF III + HSS (5:1 ratio) Pellet

Methodology

Step 5: Cooling

- Cooled at room temperature
- Solid composite formed

Step 6: Density Measurement

- Based on Archimedes principle
- ρ = Wa / (Wa − Ww)

Step 7: Observation

- Brittle behaviour observed
- Crack formation and failure

Step 8–9: Improvement

- BFS + HSS / BFS + MS used
- Reduced cracks, improved stability

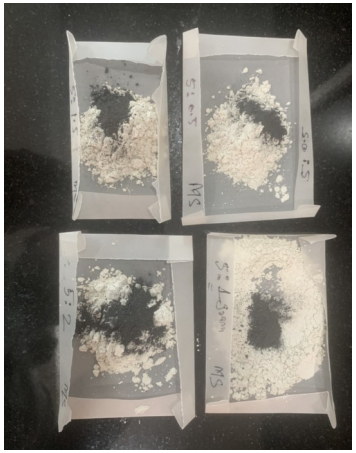


Fig 6: Before mixing
(BFS + HSS / MS)



Fig 4: Digital weighing machine



Fig 7: After mixing
(BFS + HSS / MS)

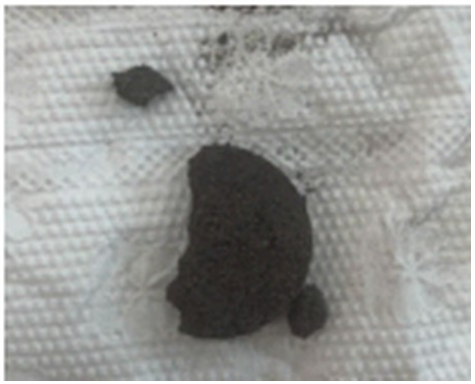


Fig 5: EAF III + HSS

Compression Test – Results & Mechanical Behaviour

Parameter	BFS–MS	BFS–HSS
Diameter	24.40 mm	23.98 mm
Thickness	4.95 mm	4.55 mm
Maximum Load	91.114 kN	**95.321 kN**
Maximum Compressive Stress	194.86 MPa	**211.06 MPa**
Displacement at Maximum Load	0.739 mm	0.687 mm
Strain at Maximum Load	0.1493	0.1510

Key observation: BFS–HSS showed a higher maximum load and ~8.3% higher compressive stress.

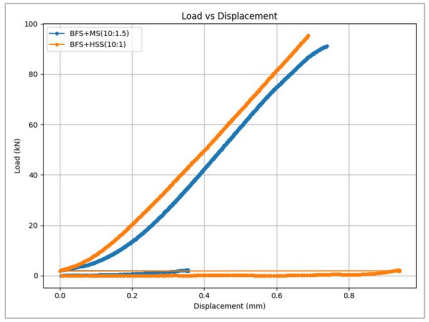


Fig 8a: Load vs Displacement

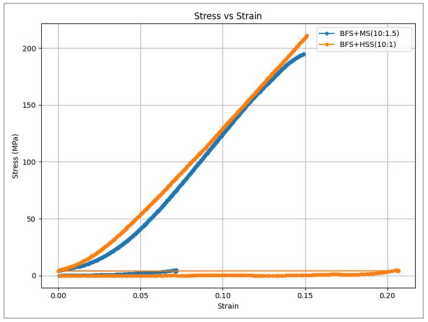


Fig 8b: Stress vs Strain

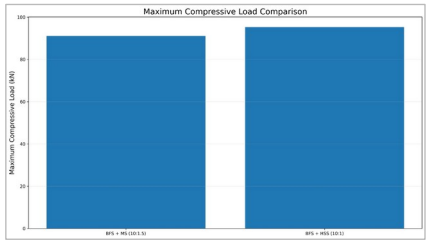


Fig 8c: Max Compressive Load

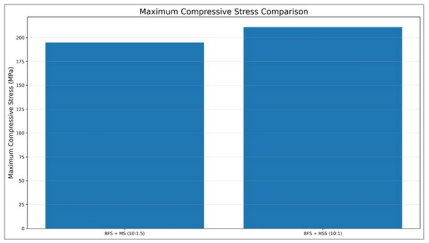
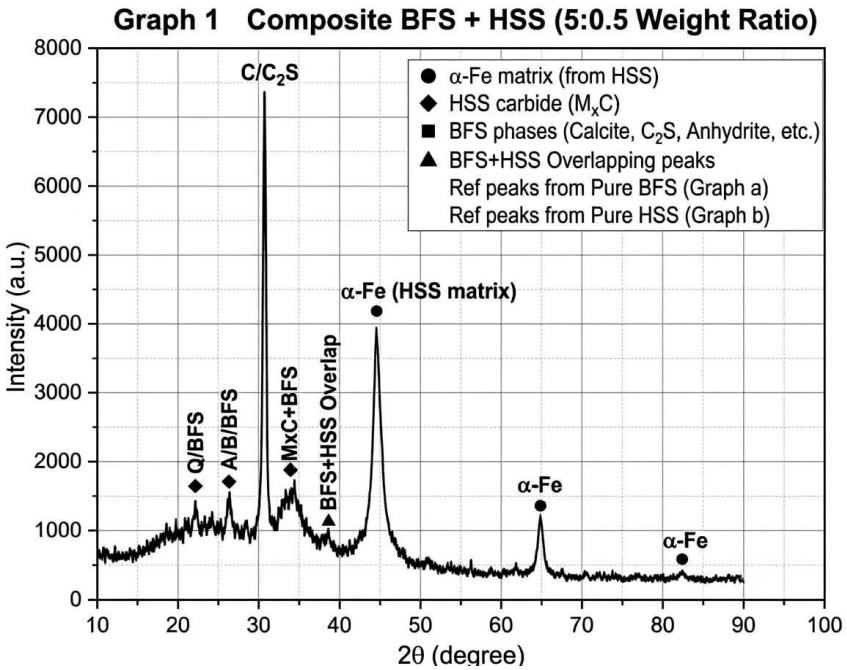
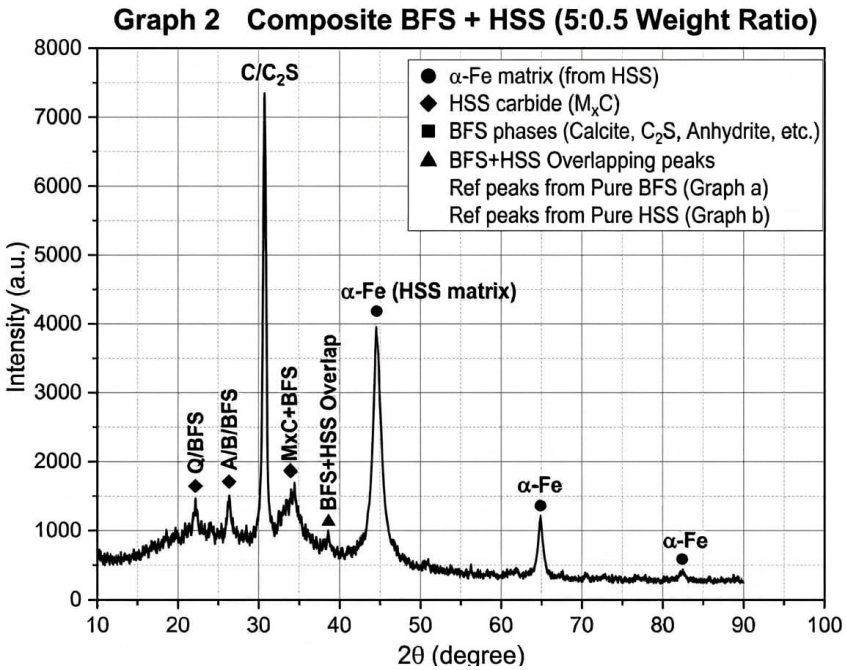


Fig 8d: Max Compressive Stress

XRD Analysis



Graph 1: XRD pattern of BFS + HSS composite



Graph 2: XRD pattern of BFS + HSS composite

Key Findings

- Phases from BFS such as calcite, C_2S, anhydrite, bassanite and quartz were identified.
- α-Fe peaks appear due to the metal reinforcement.
- The HSS composite shows peaks related to carbides.
- The MS composite has an iron-rich phase and possibly cementite-related peaks.
- Overall, the XRD patterns show that the composites are multiphase in nature.

Density Results

Sample No.	Material Composition	Composition Ratio (%)	Density (g/cm^3)
1	EAF III + HSS + PVA (binder)	EAF III (83.3%) + HSS (16.7%) + PVA (binder)	3.18
2	EAF III + MS + PVA (binder)	EAF III (83.3%) + MS (16.7%) + PVA (binder)	3.11
3	EAF III + PVA (binder)	EAF III (100%) + PVA (binder)	2.87

Standard densities

HSS = 8.15 g/cm^3

MS = 7.85 g/cm^3

BFS = 1.2 g/cm^3

Source: Internet

Potential Applications

- **Paver Tiles**

Low-cost tiles for walkways and light-traffic areas.

- **Kerb Stones & Landscaping Products**

Durable outdoor products for pathways, gardens and parks.

- **Non-Load-Bearing Blocks & Panels**

Partition walls and fillers where high strength is not required.

- **Wear-Resistant Flooring Tiles**

Abrasion-resistant tiles for workshops and industrial floors.

- **Value-Added Waste Utilization**

Converts industrial waste into useful, eco-friendly products.



Fig 9: Kerb Stones (Pathway Edging)



Fig 10: Wear-Resistant Flooring Tiles

Future Work

- **SEM–EDS Analysis**

Examine the surface morphology, particle bonding, crack formation and elemental distribution in the material.

- **Hardness Tests**

Compare the hardness of the BFS–HSS composite with the BFS–MS composite.

- **Microstructure–Property Correlation**

Combine results from SEM–EDS, XRD, density measurements and compression tests to understand material behaviour.

- **Application-Level Testing**

Make samples such as paving blocks, kerb stones and non-load-bearing panels, and test their performance in real-world conditions.



Fig 11: Scanning Electron Microscope (SEM) Setup

Conclusion

- BFS–HSS and BFS–MS hybrid composites were successfully developed with PVA as a binder.
- Heat treatment at 950 °C for 2 h improved stability of composites.
- The XRD analysis revealed that the developed composites contain multiple phases.
- Compression testing resulted in stable compressive behaviour at the tested conditions.
- The developed composites have shown promising potential for sustainable waste utilisation.

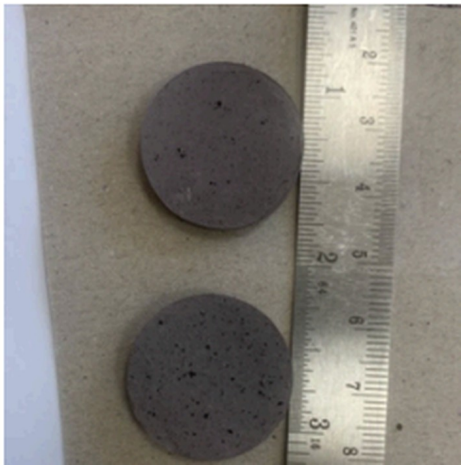


Fig 12: EAF III Based Composite

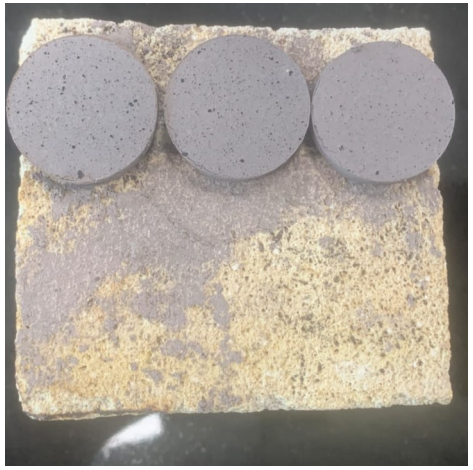


Fig 13: BFS Based Composite

References

1. H. Fu, M. A. Matthews, and L. S. Warner, “Recycling Steel from Grinding Swarf,” Waste Management, vol. 18, no. 5, pp. 321–329, 1998.

2. A. Singh, M. K. Sinha, V. Jagota, and R. Kumar, “Production, Characterisation and Utilisation of Grinding Swarf/Feedstock for Synthesis of Metal Matrix Composite through Powder Metallurgy Process,” Sādhanā, vol. 47, 2022.

Thank you!